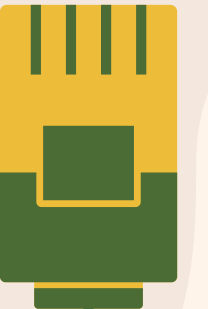
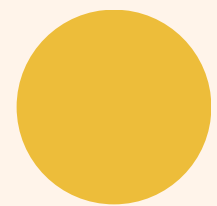


SIGNIVISION

BY SYNTAX_SQUAD





TEAM :

01

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2024CA015

02

Amisha Joshi

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Neelendra Pratap singh

2024CA060

04

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05

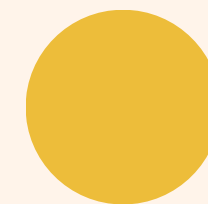
Vishal Minj

2024CA113

06

Vivek Sharma

2024CA115



Problem statement:

- There is a significant communication gap due to the general population's limited understanding of sign language.
- This gap hampers social interactions, access to services, and educational opportunities.
- A real-time sign language recognition system can bridge this gap, enabling effective communication between sign language users and non-users.

Project :

Objective :

- The project aims to develop a real-time sign language recognition model.
- The system will track hand gestures through a live camera feed and translate them into corresponding words or letters.
- The model will provide immediate text-based feedback, facilitating clear and seamless communication

Challenges Solved:

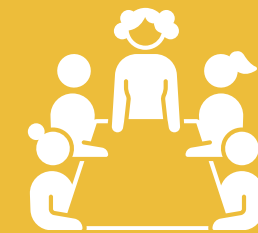


Communication Barrier:

converts sign language gestures into readable text instantly



Real-Time Translation with minimal latency



Gesture Variability:

The model is designed to accommodate variations in gesture speed, hand shapes, and orientations.

Tech Stack :

- Frontend:
 - Canvas , Lucide , React
- Uvicorn
- Backend:
 - FastAPI
- Model :
 - YoloV5 , OpenCV , Pytorch
- DataSet : labelling
- Audio Conversion (Text-to-Speech)
 - Web Speech API
- Video : webRTC



Features

01

Real-time Sign
Detection:

Detects signs in real time using a camera feed.

02

Image-based Sign
Detection:

Supports sign detection from uploaded images.

03

Real-time Video Call
Translation:

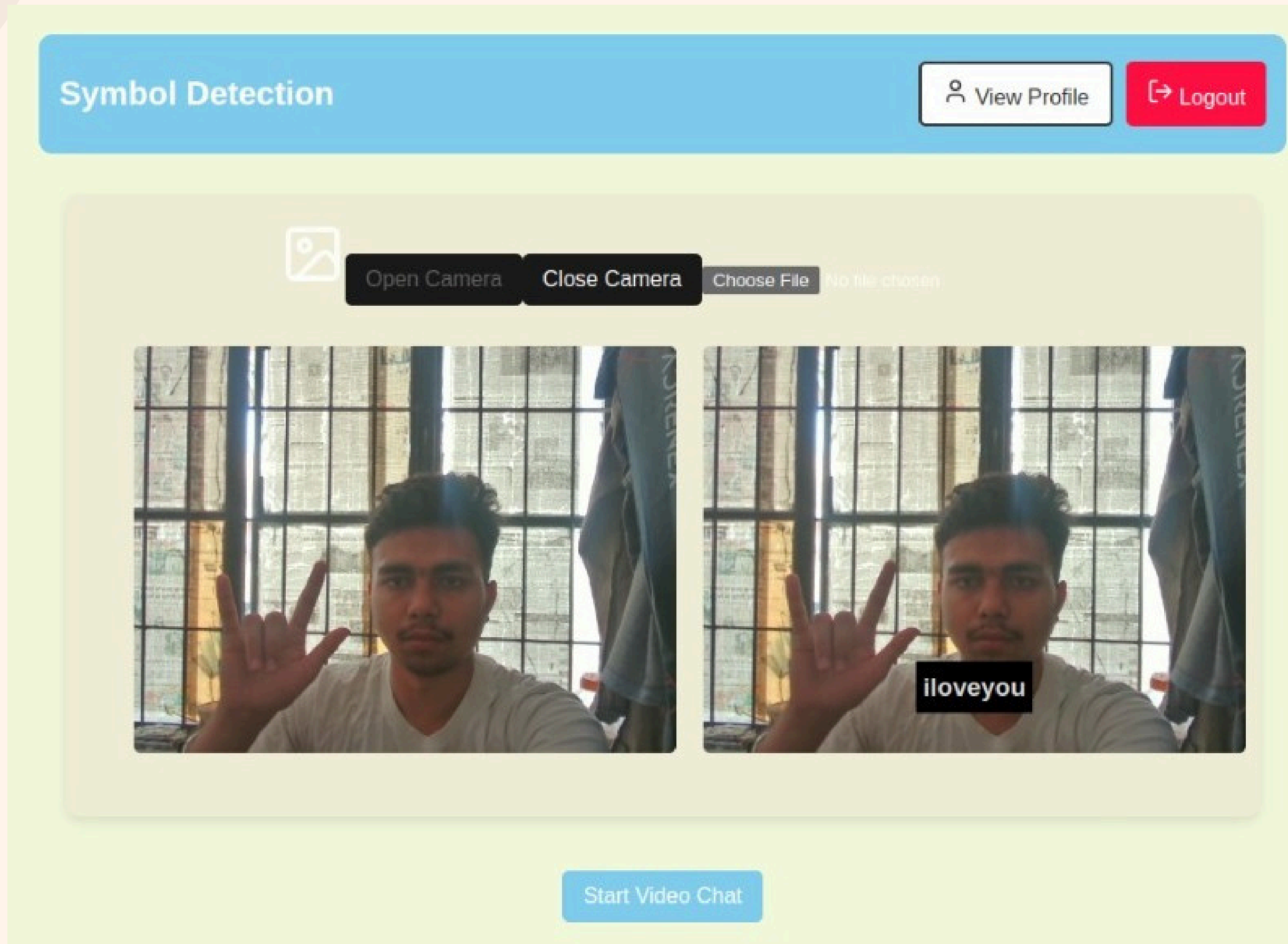
Converts the detected signs of both participants into text and audio.

04

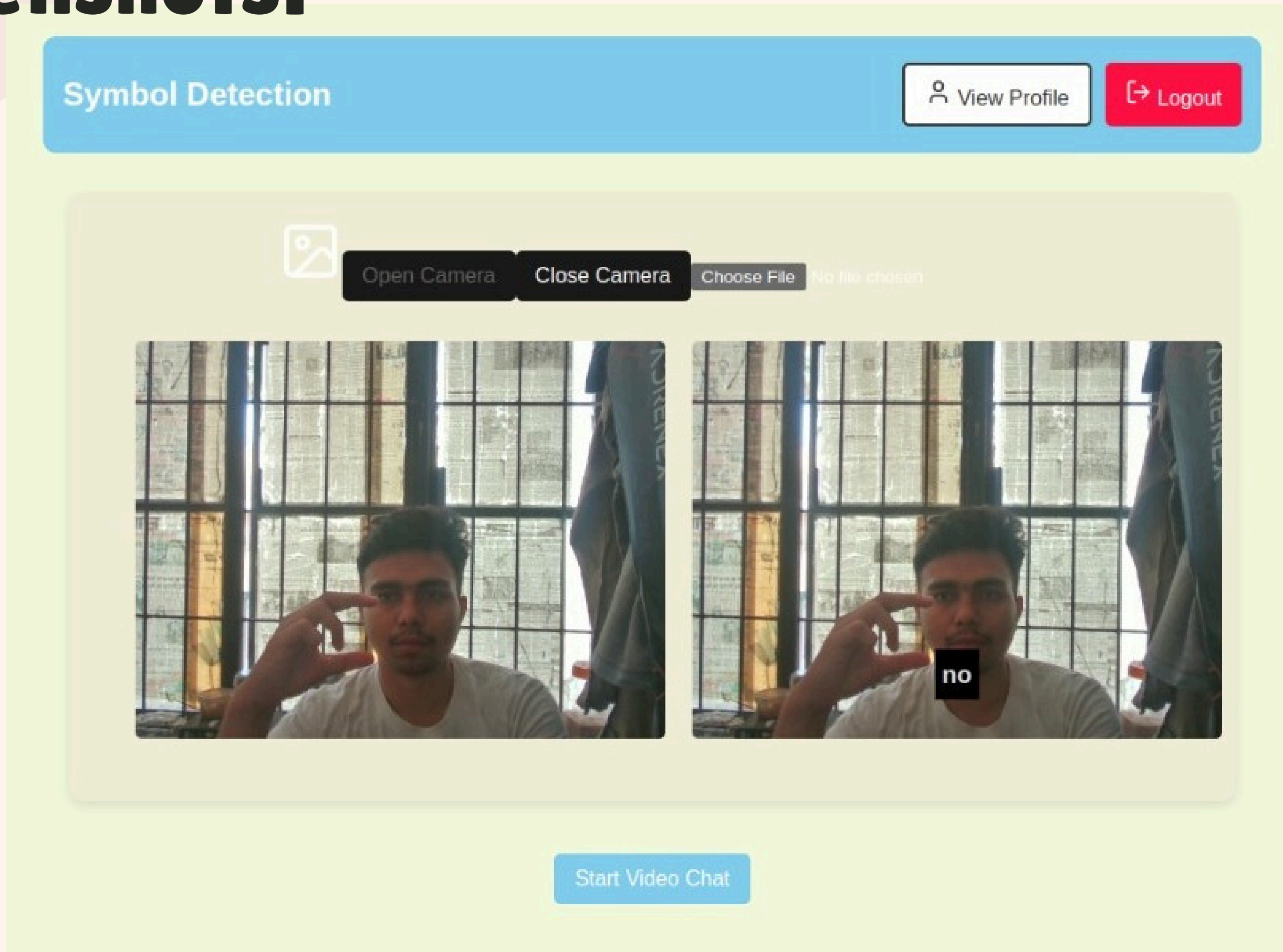
Seamless
Communication

Sends translated text and audio to the other party, breaking the language barrier.

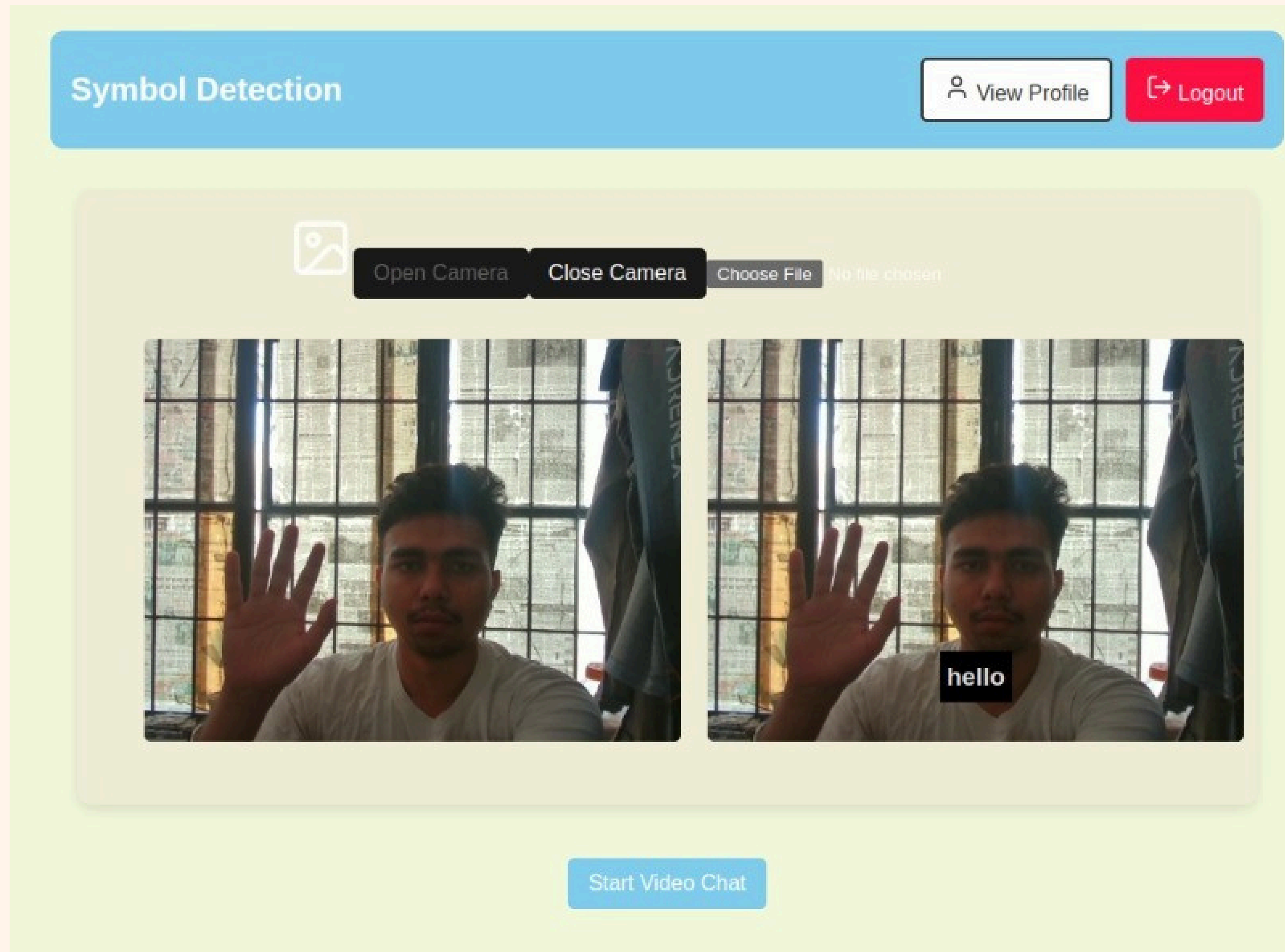
Screenshots:



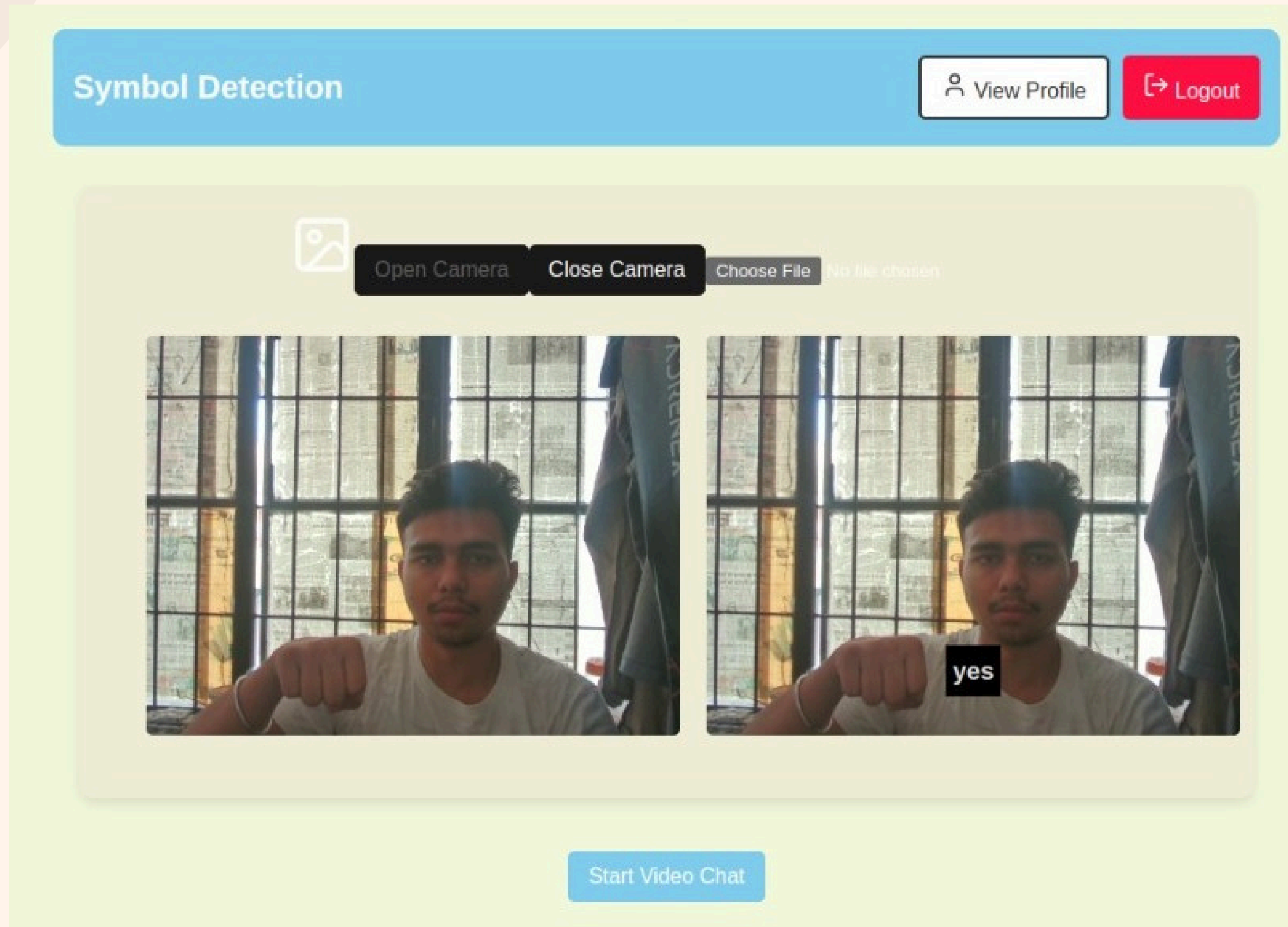
Screenshots:



Screenshots:



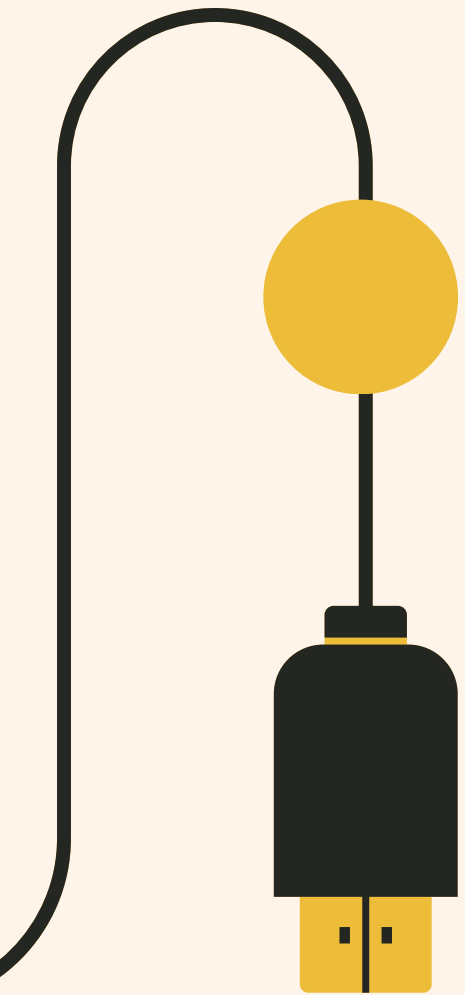
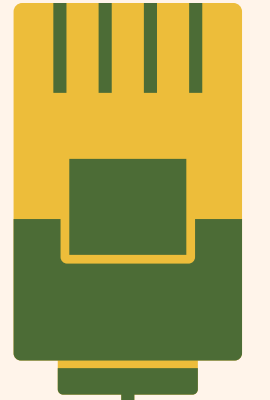
Screenshots:



Future Development



- System will incorporate text-to-speech functionality to convert recognized text into spoken audio.
- This will allow real-time video communication , bridging the gap between mute and normal person.





**THANK
YOU!**